João Victor da Silva Alves

Computer Engineering | Software Development, Web and Artificial Intelligence

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Professional Summary

Final-year Computer Engineering student specializing in Software Development, expected to graduate in December 2026. Hands-on experience in full-stack web development (PHP, JavaScript, MySQL) and applied research with Python, signal processing and machine learning, including two published papers in digital health. Analytical mindset, able to turn requirements into working, well-documented solutions. Seeking a software development internship or a junior/trainee developer position.

Education

B.Sc. in Computer Engineering — Software Development track · Universidade Ceuma, São Luís, Brazil · Jan 2023 – Dec 2026 (expected)

- Key areas: software engineering, databases, web development, signal processing and artificial intelligence.

Projects and Research

INTERMED — Medical Internship Management System · github.com/jovidasilva/Interdisciplinar-Med · *PHP, MySQL, JavaScript, HTML, CSS, Bootstrap, Composer, UML (Astah)*

- Co-developed a web system to manage medical students' clinical internships, covering student, preceptor and evaluation records.
- Owned the user registration and authentication modules and the integration between the front end and the PHP back end.
- Designed and normalized the MySQL relational database behind all system features.
- Modeled the architecture with UML diagrams and organized the codebase into modules with Composer dependency management.

ECG Segmentation for Arrhythmia Detection · *Published paper* · *Python, NumPy, SciPy, Pandas, Matplotlib, WFDB*

- Developed an algorithm for automated segmentation of P, QRS and T waves in electrocardiograms, supported by machine learning.
- Implemented the Pan-Tompkins algorithm for QRS complex detection on the MIT-BIH database, achieving over 99.3% accuracy.
- Built a preprocessing pipeline with digital filtering, noise removal and normalization of physiological signals.

AI-Powered Reconfigurable Device to Support People with ASD · *Published paper* · *Python, TensorFlow, PyTorch, Pandas, NumPy*

- Proposed an AI architecture combining CNN and LSTM networks for multimodal analysis of physiological and behavioral signals from wearable devices.
- Conducted a feasibility study of embedded (edge computing) processing for early detection of stress and sensory overload in people with Autism Spectrum Disorder.

Work Experience

Store Associate — Tenda Atacado S.A. · Itatiba, Brazil · May 2020 – Dec 2021

- Handled customer service, cash register operation and closing, merchandise checks and inventory support.
- Contributed to efficient team operations, building communication skills, accountability and attention to processes and deadlines.

Technical Skills

- **Languages:** Python, PHP, JavaScript, C, C++
- **Web development:** HTML, CSS, Bootstrap
- **Databases:** MySQL, relational data modeling
- **Data and AI:** NumPy, Pandas, SciPy, Matplotlib, TensorFlow, PyTorch, digital signal processing
- **Tools and practices:** Git, GitHub, Composer, UML (Astah)

Languages

- **Portuguese:** native
- **English:** advanced (C1)

Soft Skills

- Teamwork and clear communication in collaborative projects.
- Proactivity, organization and attention to detail.
- Fast learner of new technologies and tools.